

Examen práctico 1 Modulo 2

Angel Gonzalez - angonpad@gmail.com

1. Crear un keypair

Ya tengo un keypair creado para el examen

Key pairs (1/43) Info							Actions	Create key pair
Find Key Pair by attribute or tag							< 1 >	
☐	Name	Type	Created	Fingerprint	ID			
☐	key-pair-publicastano6	rsa	2026/02/26 18:29 GMT-5	3a:c7:60:2f:4b:47:8b:2f:46:27:1b:c2:19:72:bf:ee...	key-0d01382278456a608			
☐	claves-sebastian	rsa	2026/02/26 18:23 GMT-5	37:7b:e9:45:1a:9d:d2:64:0b:22:33:8b:61:c2:fa:62:b...	key-0627bca277900c52			
☐	JolaSecret	rsa	2026/02/19 14:38 GMT-5	fc:d6:7a:0b:6e:1d:05:5e:e7:b5:2f:67:fd:1b:1b:9c:db...	key-0c70c1cb4176c259e			
☐	aldemar/key	rsa	2026/02/20 21:06 GMT-5	6b:69:94:2c:6b:11:3a:95:6d:ee:2e:3e:21:91:62:7d:7...	key-04ceb77bcd0a880e3			
☐	juansh	rsa	2026/02/26 18:22 GMT-5	de:ec:f9:3c:31:c4:3c:35:05:47:56:72:d8:c8:90:f8:1...	key-020a1bc4c3c50e8e			
☐	lab_xbs_cron_publico	rsa	2026/02/26 18:22 GMT-5	41:d2:d9:bc:9a:8e:7e:8b:1f:bb:13:b3:d7:79:cf:6c:0a...	key-0fa5e3bb78f82a9c9			
☐	practica20	rsa	2026/02/20 21:07 GMT-5	1e:d4:ed:0a:b2:6b:0b:bb:78:2a:cd:58:6c:94:67:6e:b...	key-077efc30b3feed693			
☐	Mateo	rsa	2026/02/26 18:23 GMT-5	71:8b:9e:47:ee:57:ed:25:73:a7:c9:bb:ca:a1:cc:6e:9c...	key-0c0346dba3f85d827			
☐	keyandres	rsa	2026/02/20 21:07 GMT-5	cf:17:a7:35:e9:08:57:d5:57:3c:3e:33:4d:f1:76:cb:4d...	key-01643468b24df9508			
☐	baldo324	rsa	2026/02/26 18:26 GMT-5	57:9b:49:08:be:8c:7d:b0:48:a9:ca:a4:c9:71:6d:f3:05...	key-0659b8fbc0a2df242			
☐	landreslitz	rsa	2026/02/20 21:59 GMT-5	4c:0c:01:e9:54:55:57:7d:00:71:f5:c0:b2:9e:91:aa:6d...	key-064c894daf998e99			
☐	diana-key	rsa	2026/02/20 21:08 GMT-5	75:d9:c1:75:0e:2f:8f:34:dd:27:55:d2:15:8d:01:35:a...	key-06024c4275bb658c4			
☐	20-02-2026	rsa	2026/02/20 21:08 GMT-5	8f:56:b5:8f:35:01:9d:83:8d:32:b3:62:3f:24:9c:b1:a9...	key-091835f72777b7c9			
☐	key pair diana	rsa	2026/02/26 18:24 GMT-5	0d:e5:e0:99:3a:f5:d5:58:a0:51:1c:6f:90:37:df:28:71...	key-069e7ee2f4b0364fe			
☐	cloudwatch-test	rsa	2026/02/24 22:02 GMT-5	ae:18:de:7e:ee:83:4f:51:11:0a:d3:0c:9e:3d:4e:96:0b...	key-0f1213447f011977			
☐	kp-pedro	rsa	2026/02/26 18:26 GMT-5	0b:5e:83:bc:93:c1:6d:8c:d4:37:f8:9c:de:20:b6:1e:cf...	key-0c28a3037f982091			
☐	examen2omar	rsa	2026/02/26 18:23 GMT-5	58:a0:b3:d2:30:54:28:c2:f5:d1:06:69:2c:7a:c9:53:6f...	key-0edd565dc2a52ba1d			
☒	angelgonzalez-examen2	rsa	2026/02/26 18:25 GMT-5	a3:0f:0a:9a:60:05:32:f6:18:7e:f7:9f:a5:51:a8:6a:c0...	key-0c0438716351f6f86			
☐	SHHJohanPen	rsa	2026/02/13 19:53 GMT-5	bb:0f:11:94:9f:2b:3c:2f:38:3f:90:25:5a:3c:b9:12:53...	key-08a6209395f18d7f8			
☐	carlismodulo2	rsa	2026/02/26 18:23 GMT-5	96:36:92:2a:77:d7:15:b0:e0:f0:8d:4a:8a:5f:af:bf:57...	key-074ad7bdcf89a70a0			

2. Crear grupos de seguridad

Para la EC2

sg-0061803f241f88166 - ec2-rds-AngelGonzalezPadillaExamen2 [Actions](#)

Details

Security group name
ec2-rds-AngelGonzalezPadillaExamen2

Security group ID
sg-0061803f241f88166

Description
sg para conectarse a ssh

VPC ID
vpc-00f479057476a2db8 [v](#)

Owner
654654478122

Inbound rules count
1 Permission entry

Outbound rules count
1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1) [Manage tags](#) [Edit inbound rules](#)

☐

Name

Security group rule ID

IP version

Type

Protocol

Port range

Source

Description

☐

-

sg-06de6f930922c2645

IPv4

SSH

TCP

22

181.59.3.163/32

Linux-angel

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Outbound rules (1) [Manage tags](#) [Edit outbound rules](#)

☐

Name

Security group rule ID

IP version

Type

Protocol

Port range

Destination

Description

☐

-

sg-04955beb69da38cd29

IPv4

All TCP

TCP

0 - 65535

0.0.0.0/0

-

Para la RDS

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)
rds-ec2-AngelGonzalezPadillaExamen2

Description [Info](#)
Permite conexión segura únicamente desde el EC2

VPC [Info](#)
vpc-00f479057476a2db8 (VPC-Produccion)

Inbound rules [Info](#)

Type [Info](#)

Protocol [Info](#)

Port range [Info](#)

Source [Info](#)

Description - optional [Info](#)

MySQL/Aurora

TCP

3306

Custom

Q sg-0061803f241f88166 X EC2-asociada

Delete

MySQL/Aurora

TCP

3306

My IP

Q sg-0061803f241f88166 X 181.59.3.163/32 X

Delete

Add rule

Outbound rules [Info](#)

Type [Info](#)

Protocol [Info](#)

Port range [Info](#)

Destination [Info](#)

Description - optional [Info](#)

All traffic

All

All

Custom

Q 0.0.0.0/0 X

Delete

Add rule

Rules with destination of 0.0.0.0/0 or ::0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

X

Configuraciones

aws

Search

[Alt+S]

United States (N. Virg)

Account ID: 6546-5447-8122

anggonpad@gmail.com

Aurora and RDS

Databases

Create database

Create database

Info

Choose a database creation method

☒ Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type

Info

☐ Aurora (MySQL Compatible)


☐ Aurora (PostgreSQL Compatible)


☒ MySQL


☐ PostgreSQL


☐ MariaDB


☐ Oracle


☐ Microsoft SQL Server


☐ IBM Db2


Edition

☒ MySQL Community

Engine version

Info

View the engine versions that support the following database features.

Hide filters

☐ Show only versions that support the Multi-AZ DB cluster
Info
Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.

☐ Show only versions that support the Amazon RDS Optimized Writes
Info
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

Engine version

CloudShell

Feedback

Console Mobile App

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Privacy

Terms

Cookie preferences

Engine version

MySQL 8.4.7

☐ Enable RDS Extended Support [Info](#)
Amazon RDS Extended Support is a [paid offering](#). By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [RDS for MySQL documentation](#).

Templates

Choose a sample template to meet your use case.

☒ Production
Use defaults for high availability and fast, consistent performance.

☐ Dev/Test
This instance is intended for development use outside of a production environment.

☐ Sandbox
To develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

Availability and durability

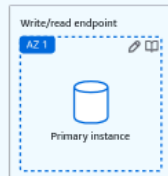
Deployment options [Info](#)

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon RDS service level agreement \(SLA\)](#).

☒ Single-AZ DB instance deployment (1 instance)

Creates a single DB instance without standby instances. This setup provides:

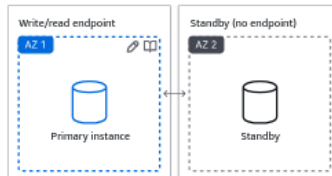
- 99.5% uptime
- No data redundancy



☐ Multi-AZ DB instance deployment (2 instances)

Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:

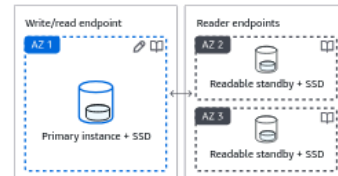
- 99.95% uptime
- Redundancy across Availability Zones



☐ Multi-AZ DB cluster deployment (3 instances)

Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:

- 99.95% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency



Settings

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

db-AngelGonzalez-Examen2

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

adminAngel

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ **Managed in AWS Secrets Manager - *most secure***
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ **Self managed**
Create your own password or have RDS create a password that you manage.

☐ Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

.....

Password strength **Very strong**

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @

Confirm master password [Info](#)

.....

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

- ☐ Show instance classes that support Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.
- ☐ Include previous generation classes
- ☐ Standard classes (includes m classes)
- ☐ Memory optimized classes (includes r and x classes)
- ☒ Burstable classes (includes t classes)

Instance type

db.t3.micro

2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Storage

Storage type [Info](#)

Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp2)

Baseline performance determined by volume size

Allocated storage [Info](#)

20

GiB

Allocated storage value must be 20 GiB to 6,144 GiB

[i](#) For high-throughput workloads, we recommend provisioning at least 100 GiB of General Purpose (SSD) storage. Lower values might result in higher latencies when the initial I/O credit balance is used up. [Learn more](#) [↗](#)

► Additional storage configuration

(Aqui ya tenía creada la EC2 así que en la consola lo voy a conectar de una vez)

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

EC2 instance [Info](#)
Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

i-01f557e1a4258ea31
EC2-AngelGonzalez-EXAMEN2

Some VPC settings can't be changed when a compute resource is added
Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group `rds-ec2-X` is added to the database and another called `ec2-rds-X` to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-00f479057476a2db8)
12 Subnets, 6 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

☐ Choose existing
Choose existing DB subnet group

☒ Automatic setup
RDS creates a new subnet group for you or reuses an existing subnet group

Nótese que agrego el SG que cree para conectarme solo desde la EC y mi IP

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

☐ Choose existing
Choose existing DB subnet group

☒ Automatic setup
RDS creates a new subnet group for you or reuses an existing subnet group

DB subnet group name
rds-ec2-db-subnet-group-1
Existing DB subnet group reused.

Public access [Info](#)
☐ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing
Choose existing VPC security groups

☐ Create new
Create new VPC security group

Additional VPC security group
Choose one or more options

rds-ec2-AngelGonzalezPadillaExamen2

Amazon RDS will add a new VPC security group `rds-ec2-2` to allow connectivity with your compute resource.

Dejamos en blanco la database ya que lo vamos a administrar desde la EC2 y desactivamos backups ya que va a ser para el examen, dejamos la encriptación habilitada

▼ Additional configuration

Database options, encryption turned on, backup turned off, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Database options

Initial database name [Info](#)

If you do not specify a database name, Amazon RDS does not create a database.

DB parameter group [Info](#)

Option group [Info](#)

Backup

☐ Enable automated backup
Creates a point-in-time snapshot of your database

Backup tags [Info](#)

Copies tags from the source database to the automated backup and snapshots respectively.

☐ Copy tags to automated backup
This is a one-time setting. Future tag modifications need manual updates.

☒ Enable encryption
Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. [Info](#)

AWS KMS key [Info](#)

✖ **Error loading KMS Keys**
User: arn:aws:iam::654654478122:user/students/anggonpad@gmail.com is not authorized to perform: kms:ListAliases on resource: * because no identity-based policy allows the kms:ListAliases action

[Diagnose with Amazon Q](#)

Maintenance

Auto minor version upgrade [Info](#)

☒ Enable auto minor version upgrade
Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see Automatically upgrading the minor engine version [documentation](#).

Maintenance window [Info](#)

Select the period you want pending modifications or maintenance applied to the database by Amazon RDS.

☐ Choose a window

☒ No preference

☐ Enable deletion protection
Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

La RDS se creo con éxito:

Successfully created database **db-angelgonzalez-examen2**

You can use settings from db-angelgonzalez-examen2 to simplify configuration of suggested database add-ons while we finish creating your DB for you.

View connection details

db-angelgonzalez-examen2

Summary

DB identifier

db-angelgonzalez-examen2

CPU

7.43%

Role

Instance

Current activity

0 Connections

Recommendations

Status

Available

Class

db.t3.micro

Engine

MySQL Community

Region & AZ

us-east-1a

Connectivity & security

Monitoring

Logs & events

Configuration

4. Creamos la EC2

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

EC2-AngelGonzalez-EXAMEN2

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debia

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-0f3caa1cf4417e51b (64-bit (x86), uefi-preferred) / ami-0bea3ccc607167c10 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260216.1 x86_64 HVM kernel-6.1

Architecture

64-bit (x...

Boot mode

uefi-preferred

AMI ID

ami-0f3caa1cf4417e51b

Publish Date

2026-02-16

Username

ec2-user

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour

☒ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

angelgonzalez-examen2

[Create new key pair](#)

▼ **Configure storage** [Info](#)

[Advanced](#)

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

[Click refresh to view backup information](#)

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

[Edit](#)

► **Advanced details** [Info](#)

▼ Summary

Number of instances | Info

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-0f3caa1cf4417e51b

Virtual server type (instance type)
t3.micro

Firewall (security group)
ec2-rds-AngelGonzalezPadillaExamen2

Storage (volumes)
1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Instance summary for i-01f557e1a4258ea31 (EC2-AngelGonzalez-EXAMEN2) Info		
Connect Instance state ▼ Actions ▼ Updated less than a minute ago		
Instance ID i-01f557e1a4258ea31	Public IPv4 address 100.55.7.75 open address ↗	Private IPv4 addresses 172.31.14.253
IPv6 address -	Instance state Running	Public DNS ec2-100-55-7-75.compute-1.amazonaws.com open address ↗
Hostname type IP name: ip-172-31-14-253.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-14-253.ec2.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	AWS Compute Optimizer finding ⊘ User: arn:aws:iam::654654478122:user/students/anggonpad@gmail.com is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action Retry
Auto-assigned IP address 100.55.7.75 [Public IP]	VPC ID vpc-00f479057476a2db8 (VPC-Produccion) ↗	Auto Scaling Group name -
IAM role -	Subnet ID subnet-00227b47a5f26ec04 ↗	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:654654478122:instance/i-01f557e1a4258ea31	
Operator -		

5. Nos conectamos al EC2 e instalamos mariadb

```
[ec2-user@ip-172-31-14-253 ~]$ mysql --version
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine
wrapper
[ec2-user@ip-172-31-14-253 ~]$
```

[0] 0:bash* "ip-172-31-14-253.ec2." 01:30 27-Feb-26

6. Nos conectamos a la RDS desde el EC2

```
[ec2-user@ip-172-31-14-253 ~]$ mysql -h db-angelgonzalez-examen2.czckyeamig9k.u
s-east-1.rds.amazonaws.com -u adminAngel -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 39
Server version: 8.4.7 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
```

[0] 0:mysql* "ip-172-31-14-253.ec2." 01:48 27-Feb-26

7. verificamos los datos

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement
.

MySQL [(none)]>CREATE DATABASE laboratorioANGELGONZALEZ
→ ;

Query OK, 1 row affected (0.021 sec)

MySQL [(none)]> SHOW DATABASES;

Database
information_schema
laboratorioANGELGONZALEZ
mysql
performance_schema
sys

5 rows in set (0.006 sec)

MySQL [(none)]>